Likelihood-ratio tests for continuous outcomes truncated by death

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Abstract

Continuous outcomes such as quality-of-life scores are often undefined after death or a related event. A common pragmatic solution is to encode such patients by the worst possible score and analyze the resulting composite outcome with a one-dimensional procedure. That preserves a clinically meaningful ordering, but it conflates treatment effects on the atom with treatment effects on the continuous part among those with observed outcomes. We propose likelihood-ratio tests for the joint null of no treatment effect on either component. The construction yields a single prespecified p-value together with interpretable component-wise effect estimates and confidence regions. We present both a parametric version and a semi-parametric version based on empirical likelihood, establish their large-sample calibration, and study their behavior in simulations. The methods are most advantageous when treatment acts primarily through one component or when the atom and continuous components move in different directions. A secondary empirical illustration with a public atom-at-zero expenditure outcome shows how the joint analysis can reveal structure that standard one-dimensional tests largely miss.

Keywords: Quality of Life, Randomized Controlled Trials,
Truncated Outcomes, Semi-Continuous Distributions, Empirical Likelihood

1 Introduction

Continuous outcomes such as quality-of-life scores, cognition scores, and days alive outside hospital are increasingly used in randomized trials of serious illness. When patients die before assessment, however, those outcomes are undefined rather than merely missing, creating the truncation-by-death problem (Frangakis and Rubin, 2002; Colantuoni et al., 2018). Composite outcomes that combine mortality with an otherwise continuous scale are common in critical care and related areas; familiar examples include ventilator-free days and days alive without life support (The COVID STEROID 2 Trial Group, 2021; Yehya et al., 2019).

One pragmatic response is to place death, or another substantively distinct event, at the bottom of a single ordered scale and analyze the resulting composite with a rank-based or related one-dimensional method. Worst-rank analyses and related composite rankings have been advocated precisely because they retain all randomized patients in the primary analysis (Lachin, 1999; McMahon and Harrell, 2000; Colantuoni et al., 2018). The same logic underlies the widespread use of atom-at-zero outcomes such as ventilator-free days, where death is assigned the worst possible value on the scale (Yehya et al., 2019). Throughout, we refer to the Mann–Whitney–Wilcoxon rank-sum test simply as the Wilcoxon test (Wilcoxon, 1945; Mann and Whitney, 1947).

That strategy is often clinically sensible, but it targets the distribution of the composite outcome itself. It does not separate treatment effects on the probability of the atom from treatment effects on the continuous outcome among patients with observed values. When the two components move in different directions, or when treatment acts mainly through one of them, a one-dimensional analysis can be attenuated because opposite component effects partially cancel on the combined scale. The issue is therefore not only one of robustness or convenience; it is also one of estimand alignment.

Other approaches address truncated-by-death outcomes from different perspectives. Survivor-only analyses condition on a post-randomization event and therefore answer a conditional question that can differ materially from the intention-to-treat comparison. Principal-stratification and survivor-average-causal-effect approaches instead target causal effects in latent survivor strata, but they require assumptions that are not identified from the observed trial data alone (Frangakis and Rubin, 2002; Colantuoni et al., 2018). Composite-endpoint approaches avoid that latent-stratum target by working directly with an ordered endpoint, but then the estimand is tied to the chosen ranking (McMahon and Harrell, 2000; Colantuoni et al., 2018). Our goal in this paper is narrower and more operational: we target the joint observed-data null of no treatment effect on the atom probability and no treatment effect on the continuous component among those for whom it is observed.

Our construction combines ideas from two-part models for semi-continuous outcomes (Aitchison, 1955; Cragg, 1971) and empirical-likelihood inference for mean restrictions (Owen, 1988, 1991; Kitamura, 2001). We develop a factorized likelihood-ratio framework that yields a single prespecified p-value for the joint null together with interpretable component-wise effect estimates and confidence regions. We present both a parametric version, which is efficient under a credible model for the positive part, and a semi-parametric version, which relaxes that distributional commitment while retaining the same Bernoulli component. The resulting procedures preserve the clinically meaningful atom used in practice, but they test a sharper question than direct one-dimensional analyses of the combined outcome.

Although we motivate the method by quality of life truncated by death, the same structure arises whenever one special value has a substantively different meaning than the remainder of an otherwise continuous scale. The target is an observed-data treatment contrast, not a principal-stratum causal estimand. The rest of the paper is organized as follows. Section 2 introduces the statistical setup and the proposed tests. Section 3 reports a simulation study, Section 4 gives a secondary empirical illustration, and Section 5 concludes. Additional simulation results and a derivation of the empirical-likelihood component are collected in Appendix A and Appendix C.

2 Method

2.1 Setup and scientific null

We observe independent and identically distributed random variables (Y_i, A_i, R_i, X_i) for $i = 1, \ldots, n$. Here Y_i is a continuous outcome when it is defined, $A_i \in \{0, 1\}$ indicates whether the outcome is defined and observed, $R_i \in \{0, 1\}$ is the treatment indicator, and X_i is a p-dimensional vector of baseline covariates. Because Y_i is only meaningful when $A_i = 1$, the scientific null hypothesis of no treatment effect is

$$\mathcal{L}(Y_i \mid A_i = 1, R_i = 1, X_i) = \mathcal{L}(Y_i \mid A_i = 1, R_i = 0, X_i),$$

$$\Pr(A_i = 1 \mid R_i = 1, X_i) = \Pr(A_i = 1 \mid R_i = 0, X_i).$$

In practice, analysts often replace this bivariate object by a single combined outcome

$$\widetilde{Y}_i = Y_i \mathbb{1}(A_i = 1) + \mathcal{E}\mathbb{1}(A_i = 0), \tag{1}$$

where $\mathcal{E}$ is a fixed atom representing the undefined outcome. The distribution of $\widetilde{Y}_i$ is therefore a mixture of a point mass at $\mathcal{E}$ and a continuous distribution on the support of $Y_i \mid A_i = 1$.

The conditional mean of the combined outcome is

$$\mathrm{E}[\widetilde{Y}_i \mid R_i, X_i] = \mathrm{E}[Y_i \mid A_i = 1, R_i, X_i] \Pr(A_i = 1 \mid R_i, X_i) + \mathcal{E} \Pr(A_i = 0 \mid R_i, X_i).$$

This quantity can be identical between treatment groups even when treatment affects both the observation probability and the continuous component. To see this, let

$$\mathrm{E}[Y_i \mid A_i = 1, R_i = 0, X_i] = \mu(X_i), \qquad \mathrm{E}[Y_i \mid A_i = 1, R_i = 1, X_i] = \mu(X_i) + \mu_\delta(X_i),$$

and, for simplicity, set $\mathcal{E} = 0$. Then

$$\begin{aligned} \Delta_{\widetilde{Y}}(X_i) &= \mathrm{E}[\widetilde{Y}_i \mid R_i = 1, X_i] - \mathrm{E}[\widetilde{Y}_i \mid R_i = 0, X_i] \\ &= \mu(X_i) \left[\Pr(A_i = 1 \mid R_i = 1, X_i) - \Pr(A_i = 1 \mid R_i = 0, X_i)\right] \\ &\quad + \mu_\delta(X_i) \Pr(A_i = 1 \mid R_i = 1, X_i). \end{aligned} \tag{2}$$

Choosing

$$\mu_\delta(X_i) = \mu(X_i) \left[\Pr(A_i = 1 \mid R_i = 0, X_i) \Pr(A_i = 1 \mid R_i = 1, X_i)^{-1} - 1\right]$$

forces $\Delta_{\widetilde{Y}}(X_i) = 0$ even though both components depend on treatment. Thus testing $\Delta_{\widetilde{Y}}(X_i) = 0$ is generally not equivalent to testing no treatment effect on the joint observed-data structure, and the latter requires two restrictions.

Throughout, $\mathcal{E}$ is treated as a fixed, prespecified atom representing the clinically worst value on the combined scale. The event $A_i = 0$ therefore denotes the substantive atom event, such as death before assessment or zero days alive out of hospital, rather than ordinary measurement missingness. Our estimands are defined with respect to the observed-data law of $(A_i, Y_i \mid A_i = 1)$, so identifiability is with respect to that observed-data structure rather than a latent principal stratum. The inferential development below assumes positive probabilities of $A_i = 1$ in both treatment groups, identifiable finite-dimensional models for the Bernoulli component, and, when the parametric procedure is used, a correctly specified model for the conditional outcome distribution.

To exploit that structure, we model the binary and continuous components separately. Let

$$\begin{aligned} A_i \mid R_i, X_i &\sim \text{Bernoulli}\left(\pi_i\right), \\ Y_i \mid A_i = 1, R_i, X_i &\sim F\left(\cdot; \mu_i, \Psi\right), \end{aligned}$$

where $\pi_i = \Pr(A_i = 1 \mid R_i, X_i)$ is the observation probability, $\mu_i = \mathrm{E}[Y_i \mid A_i = 1, R_i, X_i]$ is a mean parameter for the observed outcomes, and Ψ collects nuisance parameters governing the distribution of the continuous component. If $f(\cdot; \mu_i, \Psi)$ denotes the corresponding density, the observed-data likelihood is

$$L_n(\mu_i, \pi_i, \Psi) = \prod_{i=1}^{n} f\left(Y_i; \mu_i, \Psi\right)^{A_i} \pi_i^{A_i} \left(1 - \pi_i\right)^{1 - A_i}.$$

This factorizes as

$$L_n(\mu_i, \pi_i, \Psi) = L_{Y,n}(\mu_i, \Psi) L_{A,n}(\pi_i),$$

where

$$L_{Y,n}(\mu_i, \Psi) = \prod_{i=1}^{n} f\left(Y_i; \mu_i, \Psi\right)^{A_i},$$
$$L_{A,n}(\pi_i) = \prod_{i=1}^{n} \pi_i^{A_i} \left(1 - \pi_i\right)^{1-A_i}.$$

The factorization follows from the likelihood construction; it does not require independence between the value of Y_i and the event $A_i = 1$.

We now impose generalized linear models on the two components:

$$\operatorname{logit} \Pr(A_i = 1 \mid R_i, X_i) = \beta_0 + \beta_\delta R_i + s_\beta(X_i), \tag{3}$$
$$\mathrm{E}[Y_i \mid A_i = 1, R_i, X_i] = \mu_0 + \mu_\delta R_i + s_\mu(X_i), \tag{4}$$

where s_μ and s_β are finite-dimensional regression terms for the additional covariates. The parameter μ_δ is the mean difference among observed outcomes, and β_δ is the log odds-ratio of being observed.

On the original outcome scale, μ_δ quantifies how treatment shifts the positive part of the outcome distribution among subjects with observed outcomes, whereas β_δ quantifies how treatment changes the odds that the outcome is observed at all. Reporting the pair therefore separates improvement in the continuous component from improvement in the probability of escaping the atom. This two-component contrast is the scientific target of the proposed likelihood-ratio test. It remains informative even when a single combined-outcome contrast such as $\Delta_{\widetilde{Y}}(X_i)$ is zero because the two components offset each other.

For fixed treatment contrasts $(\mu_\delta, \beta_\delta)$, the profile likelihood-ratio statistic is

$$\begin{aligned}
W_n^P(\mu_\delta, \beta_\delta) &= W_{Y,n}^P(\mu_\delta) + W_{A,n}^P(\beta_\delta), \\
W_{Y,n}^P(\mu_\delta) &= -2 \log \frac{\sup\limits_{\mu_0, s_\mu, \Psi} L_{Y,n}(\mu_0, \mu_\delta, s_\mu, \Psi)}{\sup\limits_{\mu_0, \mu_\delta, s_\mu, \Psi} L_{Y,n}(\mu_0, \mu_\delta, s_\mu, \Psi)}, \\
W_{A,n}^P(\beta_\delta) &= -2 \log \frac{\sup\limits_{\beta_0, s_\beta} L_{A,n}(\beta_0, \beta_\delta, s_\beta)}{\sup\limits_{\beta_0, \beta_\delta, s_\beta} L_{A,n}(\beta_0, \beta_\delta, s_\beta)}.
\end{aligned} \tag{5}$$

The null hypothesis of no treatment effect is tested with $W_n^P(0, 0)$.

2.2 Parametric likelihood-ratio test

In the parametric version, one specifies a parametric model for the observed outcomes. For example, combining Equation (4) with a normal model yields

$$Y_i \mid A_i = 1, R_i, X_i \sim N\left(\mu_0 + \mu_\delta R_i + s_\mu(X_i), \sigma^2\right).$$

This produces the usual Gaussian likelihood-ratio statistic for $W_{Y,n}^P(\mu_\delta)$. Other parametric families can be used in the same way when scientifically appropriate.

Proposition 1. *Assume that the parametric model for $Y_i \mid A_i = 1, R_i, X_i$ is correctly specified, that the regression models in Equations (3) and (4) are finite-dimensional and correctly specified, that the true parameter lies in the interior of the parameter space, that the model is identifiable with nonsingular Fisher information, and that $\Pr(A_i = 1 \mid R_i = r, X_i) > 0$ almost surely for $r = 0, 1$. Let $n_r = \sum_{i=1}^{n} \mathbb{1}(R_i = r)$ and assume $n_0, n_1 \to \infty$ with $n_1/n_0 \to c \in (0, \infty)$. Then, under the null hypothesis $\mu_\delta = \beta_\delta = 0$, the profile likelihood-ratio statistic $W_n^P(0, 0)$ in Equation (5) converges in distribution to χ^2_2.*

Proof. Under the stated regularity conditions, the joint likelihood is a regular finite-dimensional likelihood with two scalar restrictions under the null: one in the continuous component and one in the Bernoulli component. Because the likelihood factorizes into $L_{Y,n}$ and $L_{A,n}$, the joint likelihood-ratio statistic is exactly the sum of the corresponding component likelihood-ratio statistics. Wilks' theorem (Wilks, 1938) therefore yields the χ^2_2 limit. □

In practice, this test is easy to compute with standard software: one fits the observed-outcome model with and without the treatment term and does the same for the logistic model for A_i, then adds the two log-likelihood differences according to Equation (5). At the 5% level the critical value is 5.99, and at the 1% level it is 9.21.

2.3 Semi-parametric likelihood-ratio test

The parametric test depends on the chosen distributional model for the observed outcomes. To relax that assumption, we replace the first component by a profile empirical likelihood while retaining the Bernoulli likelihood for A_i. When there are no additional covariates, the resulting procedure is fully non-parametric for the observed-outcome component. Covariate-adjusted variants can be built by combining the same logistic model for A_i with a regression empirical likelihood for μ_δ, but the no-covariate case is enough to display the main construction.

To simplify the presentation, we now assume that X_i is absent. Let

$$\mathcal{I}_0 = \{i = 1, \ldots, n : R_i = 0, A_i = 1\}, \qquad \mathcal{I}_1 = \{i = 1, \ldots, n : R_i = 1, A_i = 1\}.$$

Lemma 3 in Appendix C shows that, for fixed (μ_0, μ_δ), the empirical-likelihood deviance for the observed outcomes is

$$\widetilde{W}_{Y,n}(\mu_0, \mu_\delta) = 2\left(\sum_{i \in \mathcal{I}_0} \log\left(1 + \lambda_0(Y_i - \mu_0)\right) + \sum_{j \in \mathcal{I}_1} \log\left(1 + \lambda_1(Y_j - \mu_0 - \mu_\delta)\right)\right), \tag{6}$$

where λ_0 and λ_1 solve

$$\frac{1}{|\mathcal{I}_0|} \sum_{i \in \mathcal{I}_0} \frac{Y_i - \mu_0}{1 + \lambda_0(Y_i - \mu_0)} = 0,$$
$$\frac{1}{|\mathcal{I}_1|} \sum_{j \in \mathcal{I}_1} \frac{Y_j - \mu_0 - \mu_\delta}{1 + \lambda_1(Y_j - \mu_0 - \mu_\delta)} = 0.$$

The profile version is obtained by minimizing over the nuisance mean:

$$\widetilde{W}^P_{Y,n}(\mu_\delta) = \inf_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta).$$

The admissible values of μ_0 are those for which the corresponding constrained means lie in the convex hull of the observed outcomes in each group. For interior feasible means, the dual parameters λ_0 and λ_1 are determined by one-dimensional root equations with unique roots on their admissible intervals, and the profiled nuisance mean is then minimized over its feasible set. If a candidate constraint falls outside the convex-hull region, the corresponding empirical-likelihood statistic is treated as infinite.

The semi-parametric test statistic is then

$$\widetilde{W}_n^P(\mu_\delta, \beta_\delta) = \widetilde{W}_{Y,n}^P(\mu_\delta) + W_{A,n}^P(\beta_\delta). \tag{7}$$

Proposition 2. *Assume that there are no additional covariates, that* $\Pr(A_i = 1 \mid R_i = r) > 0$ *for* $r = 0, 1$*, and that* $n_0, n_1 \to \infty$ *with* $n_1/n_0 \to c \in (0, \infty)$*. Suppose further that the Bernoulli model for* A_i *is correctly specified, that under the null hypothesis the observed outcomes in the two treatment groups have a common finite mean, strictly positive finite variance, and support whose convex hull contains that mean as an interior point. Then, under* $\mu_\delta = \beta_\delta = 0$*, the semi-parametric statistic* $\widetilde{W}_n^P(0, 0)$ *in Equation (7) converges in distribution to* χ_2^2*.*

Proof. Under the stated conditions, $W_{A,n}^P(0)$ is the usual one-parameter Bernoulli likelihood-ratio statistic, so Wilks' theorem gives a χ_1^2 limit. For the observed outcomes, standard two-sample empirical-likelihood theory for a mean-difference constraint implies that $\widetilde{W}_{Y,n}^P(0)$ has the Wilks-type χ_1^2 limit; see Owen (1991). The joint criterion is the sum of these two orthogonal components, so the overall null statistic has the χ_2^2 limit. □

2.4 Inference and interpretation

Confidence regions follow by inverting the corresponding likelihood-ratio statistic. For the parametric procedure, an asymptotic $(1-\alpha)100\%$ simultaneous confidence region for $(\mu_\delta, \beta_\delta)$ is

$$\mathrm{CI}_{1-\alpha}^{(\mu_\delta, \beta_\delta)} = \left\{(m, b) \in \mathbb{R}^2 : W_n^P(m, b) \le \chi_2^2(1-\alpha)\right\}.$$

For the semi-parametric procedure, one replaces W_n^P by $\widetilde{W}_n^P$. Correspondingly, the marginal intervals are

$$\begin{aligned}
\mathrm{CI}_{1-\alpha}^{\mu_\delta} &= \left\{m \in \mathbb{R} : W_{Y,n}^P(m) \le \chi_1^2(1-\alpha)\right\} \quad \text{or} \quad \left\{m \in \mathbb{R} : \widetilde{W}_{Y,n}^P(m) \le \chi_1^2(1-\alpha)\right\}, \\
\mathrm{CI}_{1-\alpha}^{\beta_\delta} &= \left\{b \in \mathbb{R} : W_{A,n}^P(b) \le \chi_1^2(1-\alpha)\right\}.
\end{aligned}$$

We recommend reporting the simultaneous confidence region because it displays the joint uncertainty about the effect on the continuous and binary components.

Because many applied reports still request summaries on the combined-outcome scale Δ, one can also project or profile the likelihood ratio onto Δ. We regard those intervals as secondary because they collapse the two-component effect back to a single scale; Appendix B describes the corresponding profile and projected constructions.

3 Simulation study

The simulation study had two purposes: to assess null calibration and to compare power with standard one-dimensional analyses when treatment acts through the atom, the positive part, or both. In every replicate, the two treatment groups had equal size, the atom was fixed at $\mathcal{E} = 0$, and each subject contributed the combined outcome $\widetilde{Y}_i = A_i Y_i$, where $A_i \mid R_i = r$ was Bernoulli with scenario-specific probability π_r and $Y_i \mid (A_i = 1, R_i = r)$ followed the corresponding survivor distribution.

Table 1 lists the data-generating mechanisms. The scenarios were chosen to isolate the practically important comparisons: a low-atom normal reference case (Scenario 1), a pure atom-probability effect (Scenario 2), an antagonistic case in which the atom and continuous components move in opposite directions while the mean of $\widetilde{Y}$ stays fixed (Scenario 3), a concordant joint improvement case (Scenario 4), a gamma shape-change case with survivor-mean shift (Scenario 5), and a rare-outlier contamination case that is deliberately favorable to rank-based methods (Scenario 6).

Table 1: Simulation scenarios for the TruncComp2 power study

Scenario	Configuration	$P(A = 1 \mid R)$	$Y \mid A = 1, R$
S1	Normal survivor shift, low atom	$\pi_0 = \pi_1 = 0.85$	$Y \mid A = 1, R = 0 \sim N(3, 1)$, $Y \mid A = 1, R = 1 \sim N(3 + \delta_h, 1)$, $\delta_h \in \{0, 0.20, 0.35, 0.50\}$
S2	Survival-only effect	$\pi_0 = 0.45$, $\pi_1 \in \{0.45, 0.50, 0.55, 0.60\}$	$Y \mid A = 1, R = 0 \sim Y \mid A = 1, R = 1 \sim N(3, 1)$
S3	Antagonistic cancellation	$\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$	$Y \mid A = 1, R = 0 \sim N(3, 1)$, $Y \mid A = 1, R = 1 \sim N(1.5/\pi_1, 1)$
S4	Concordant joint improvement	$\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$	$Y \mid A = 1, R = 0 \sim N(3, 1)$, $Y \mid A = 1, R = 1 \sim N(3 + 0.15h, 1)$
S5	Gamma shape-change with mean shift	$\pi_0 = \pi_1 = 0.65$	$Y \mid A = 1, R = 0 \sim \Gamma(9, 3/9)$, $Y \mid A = 1, R = 1 \sim \Gamma(k_h, m_h/k_h)$, $(k_h, m_h) \in \{(9, 3.0), (6, 3.2), (4, 3.4), (2.5, 3.6)\}$
S6	Rare-outlier contamination	$\pi_0 = \pi_1 = 0.80$	$Y \mid A = 1, R = r$ is $0.95N(3 + \delta_h r, 0.35^2) + 0.05N(10 + \delta_h r, 0.5^2)$, $\delta_h \in \{0, 0.10, 0.20, 0.35\}$

For each scenario we considered effect levels $h \in \{0, 1, 2, 3\}$, with $h = 0$ as the null configuration, and sample sizes from 50 to 500 observations per group in increments of 50. This yielded $6 \times 4 \times 10 = 240$ scenario-effect-sample-size cells, and each cell used 10,000 Monte Carlo repetitions. Power and empirical size were estimated as rejection proportions at the 5% level, so the worst-case Monte Carlo standard error of an estimated rejection rate was approximately 0.005. We compared the parametric and semi-parametric likelihood-ratio tests with Welch's two-sample t-test and the Wilcoxon rank-sum test applied directly to the combined outcome; Appendix A records the exact simulation protocol and implementation details needed to reproduce the study.

Figure 1 shows power as a function of sample size at the strongest non-null level for the four scenarios that most sharply separate the methods: the survival-only case (Scenario 2), the antagonistic cancellation case (Scenario 3), the gamma misspecification case (Scenario 5), and the rare-outlier contamination case (Scenario 6). Figure 2 complements that display by plotting power as a function of the effect level at $n = 250$ observations per group for the same four scenarios. The supplementary material reports the corresponding power curves for Scenarios 1 and 4, the full scenario-specific power tables for all 240 cells, and the null-calibration displays for the $h = 0$ settings.

Three patterns are most important. First, in Scenario 2, where treatment acts only on the atom probability, the proposed likelihood-ratio procedures gain power because they target the Bernoulli component directly, whereas the t-test and Wilcoxon test must infer that effect indirectly through the combined outcome. Second, Scenario 3 is the identifying example for the paper: the mean of the combined outcome is held fixed even though both components change. In that antagonistic setting, the likelihood-ratio procedures continue to detect the treatment effect, whereas the one-dimensional tests are greatly attenuated because they target a contrast that has been constructed to cancel.

The remaining scenarios show the limits of any single preferred procedure. Scenario 5 narrows

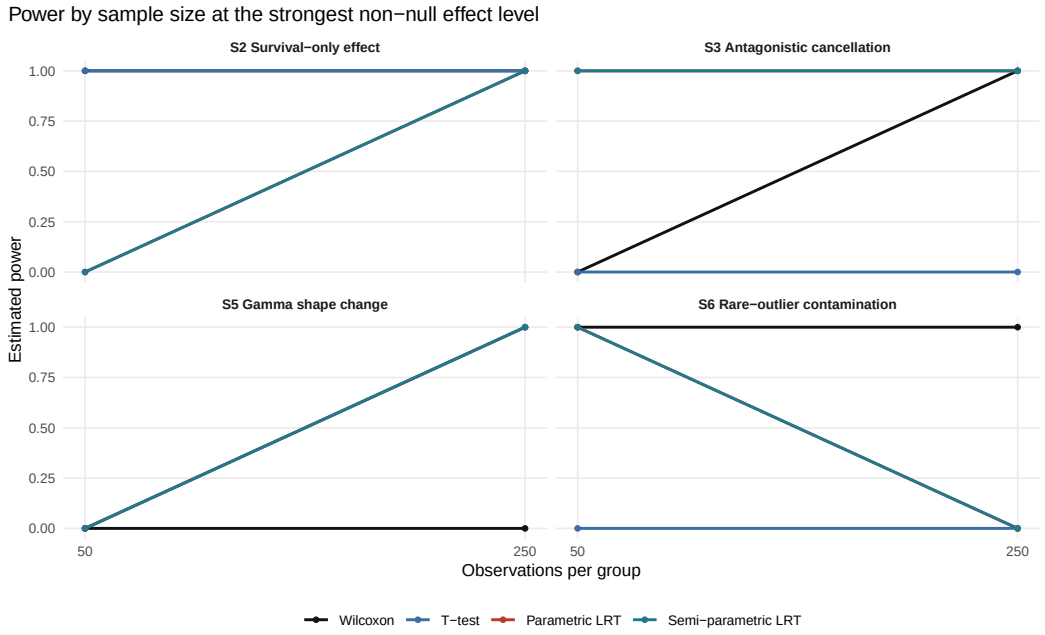

Figure 1: Estimated power as a function of sample size at the strongest non-null level for Scenarios 2, 3, 5, and 6.

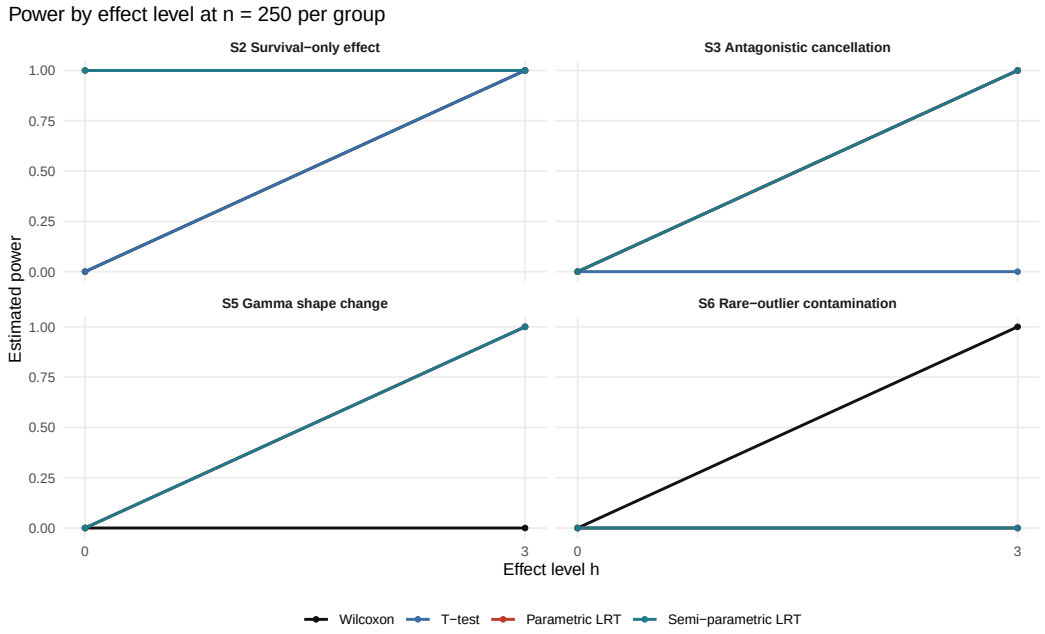

Figure 2: Estimated power as a function of effect level at $n = 250$ observations per group for Scenarios 2, 3, 5, and 6.

the gap between the parametric and semi-parametric analyses because the positive part changes shape as well as mean, and Scenario 6 is deliberately favorable to rank-based methods because the positive part contains rare high outliers. Across these settings, however, the two proposed tests are strongest precisely when the data reflect the two-component structure they are designed to detect.

The supplementary null-calibration display in Figure 5 shows that all four procedures remain close to the nominal 5% level in the null settings we considered. The supplementary figures and long tables document the full scenario-by-sample-size patterns, including the reference Scenario 1 and the concordant-improvement Scenario 4.

4 Empirical illustration

As a secondary empirical illustration, we use the public RAND Health Insurance Experiment year-1 comparison between the 0% and 50% coinsurance plans ($n = 3,462$). Although this is not a truncation-by-death data set, it has the same atom-plus-continuous structure: Year 1 inpatient expenditure (USD) has a point mass at zero corresponding to no inpatient expenditure, with a positive continuous part among subjects with nonzero outcomes.

Figure 3 shows the empirical distributions in the two treatment groups together with the simultaneous confidence region for the semi-parametric analysis. In our notation, we code $A = 0$ when the outcome equals zero and $A = 1$ otherwise; Y then denotes the positive part of the outcome among subjects with $A = 1$.

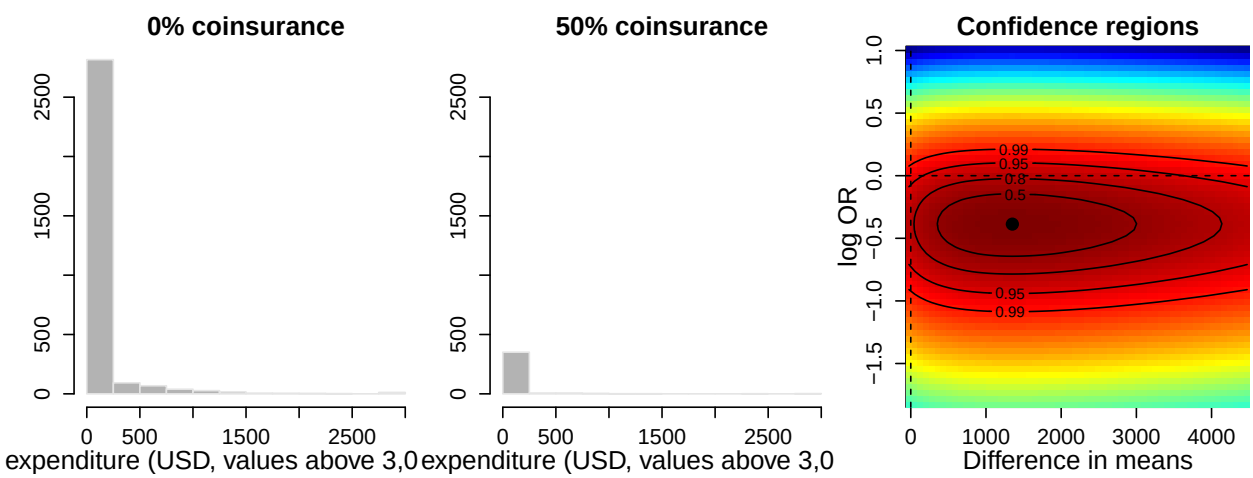

Figure 3: the RAND Health Insurance Experiment: histograms of year 1 inpatient expenditure by treatment group (left and middle panels) and the semi-parametric simultaneous confidence region for $(\mu_\delta, \beta_\delta)$ (right panel).

Standard one-dimensional analyses provide only weak evidence of a difference: Welch's two-sample t-test yields p-value 0.418, and the Wilcoxon test yields p-value 0.067. By contrast, the parametric likelihood-ratio test yields p-value 0.001, and the semi-parametric analysis yields p-value 0.027. The semi-parametric estimate of the positive-part mean difference is 1,350.9, while the fitted odds ratio for escaping the atom is 0.679.

This contrast is substantively informative because The fitted components point in opposite direc-

tions: relative to 0% coinsurance, 50% coinsurance has larger positive-part means but lower odds of having positive inpatient expenditure. The illustration is therefore not presented as a definitive clinical truncation-by-death analysis, but as a public example of how the proposed joint test can reveal a two-component signal that is muted on the raw combined-outcome scale.

5 Discussion

We have proposed parametric and semi-parametric likelihood-ratio tests for continuous outcomes with a clinically meaningful atom. The key step is to retain the atomized outcome used in practice while testing the joint null of no treatment effect on the atom probability and no treatment effect on the positive part. This yields one prespecified p-value together with interpretable component-wise effect estimates and confidence regions.

The simulations show why that distinction matters. The gain is largest when treatment acts mainly through one component or when the atom and positive part move in opposite directions, because those are exactly the settings in which a one-dimensional composite can dilute or cancel signal. The empirical illustration serves the same purpose in a public data set: the joint tests detect structure that is only weakly visible to standard analyses of the raw combined outcome.

This observed-data focus should also guide interpretation. The parameter pair $(\mu_\delta, \beta_\delta)$ is not a survivor-average causal effect or a principal-stratum estimand; it describes the two components of the observed outcome distribution under randomization. The procedures are asymptotic, require positive numbers of observed outcomes in both groups, and the parametric version is only as credible as the model specified for $Y \mid A = 1$. The semi-parametric version relaxes that distributional assumption, but it still relies on regularity of the Bernoulli model and on feasible empirical-likelihood constraints.

For trials that require a single prespecified primary analysis, the framework offers one joint p-value together with clinically interpretable follow-up summaries for the binary and continuous components. That makes it attractive when investigators want to preserve the meaning of a worst-rank or atom-at-zero endpoint without collapsing the scientific question to a single distribution on the combined scale.

An implementation of both procedures, including the code used to generate the manuscript analyses, is available in the R package `TruncComp2` on GitHub (Jensen and Lange, 2018).

A Supplementary simulation results

This appendix collects the simulation figures and tables referenced in Section 3. It also records the exact simulation protocol implemented in `simulation-study/scripts/run-simulation-study.R` and `simulation-study/R/simulation-study.R`.

The simulation study used the full factorial design generated by the code defaults: six scenarios, four effect levels $h \in \{0, 1, 2, 3\}$, and ten per-group sample sizes $n \in \{50, 100, \ldots, 500\}$, for a total of 240 cells. Each replicate contained exactly n control observations and n treatment observations. Within group $r \in \{0, 1\}$, the indicator A of whether the continuous outcome was observed was drawn as $A \sim \text{Bernoulli}(\pi_r)$. Conditional on $A = 1$, the positive outcome was drawn from a scenario-specific distribution $F_{r,h}$; conditional on $A = 0$, the observed combined outcome was set to the atom 0. Thus the analyzed endpoint was always $\widetilde{Y} = AY$, and no covariate adjustment was used.

The six scenario families were as follows. Scenario 1 used $\pi_0 = \pi_1 = 0.85$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(3 + \delta_h, 1)$ with $\delta_h \in \{0, 0.20, 0.35, 0.50\}$. Scenario 2 used $\pi_0 = 0.45$, $\pi_1 \in \{0.45, 0.50, 0.55, 0.60\}$, and $Y \mid A = 1, R = 0 \sim Y \mid A = 1, R = 1 \sim N(3, 1)$. Scenario 3 used $\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(1.5/\pi_1, 1)$, so that $E(\widetilde{Y} \mid R = 1) = E(\widetilde{Y} \mid R = 0) = 1.5$ at every effect level even though both components change. Scenario 4 used $\pi_0 = 0.50$, $\pi_1 \in \{0.50, 0.55, 0.60, 0.65\}$, $Y \mid A = 1, R = 0 \sim N(3, 1)$, and $Y \mid A = 1, R = 1 \sim N(3 + 0.15h, 1)$. Scenario 5 used $\pi_0 = \pi_1 = 0.65$, $Y \mid A = 1, R = 0 \sim \Gamma(9, 3/9)$, and $Y \mid A = 1, R = 1 \sim \Gamma(k_h, m_h/k_h)$ with $(k_h, m_h) \in \{(9, 3.0), (6, 3.2), (4, 3.4), (2.5, 3.6)\}$, where $\Gamma(k, \theta)$ denotes a gamma distribution with shape k and scale θ. Scenario 6 used $\pi_0 = \pi_1 = 0.80$ and $Y \mid A = 1, R = r \sim 0.95\, N(3 + \delta_h r, 0.35^2) + 0.05\, N(10 + \delta_h r, 0.5^2)$ with $\delta_h \in \{0, 0.10, 0.20, 0.35\}$.

For each cell, the driver called `set.seed(20260416 + cell_id - 1)` before generating the 10,000 replicates, where `cell_id` increases first over scenarios, then over effect levels, and then over sample sizes. Each replicate was analyzed by four methods applied to the combined outcome data frame produced by `simulateTruncatedData()`: `wilcox.test(Y ~ R, exact = FALSE)` for the Wilcoxon rank-sum test, `t.test(Y ~ R)` for Welch's two-sample t-test, `truncComp(Y ~ R, atom = 0, method = "LRT")` for the parametric likelihood-ratio test, and `truncComp(Y ~ R, atom = 0, method = "SPLRT")` for the semi-parametric likelihood-ratio test. In the package implementation, the parametric test combines a logistic likelihood-ratio comparison for A with a Gaussian likelihood-ratio comparison for Y among observations with $A = 1$, whereas the semi-parametric test combines the same logistic component with an empirical-likelihood test for the survivor mean difference. Both proposed procedures report a two-degree-of-freedom χ^2-calibrated joint p-value.

Within each cell, rejection rates were computed as the proportion of successful replicates with p-value below 0.05 for each method, and the Monte Carlo standard error was computed from that rejection proportion using the number of successful replicates for the given method. If a method call errored or returned a non-finite p-value, that replicate was counted as a failure for that method. The aggregated results object stores method-specific failure rates and runtimes, and the supplementary tables report the largest failure rate across the four methods in each cell. For the manuscript displays, Figure 1 uses the strongest non-null effect level ($h = 3$) for Scenarios 2, 3, 5, and 6; Figure 2 uses all four effect levels at $n = 250$ per group for those same scenarios; Figure 4 uses $h = 3$ for Scenarios 1 and 4; and Figure 5 uses the null cells with $h = 0$ across all six scenarios. The six long supplementary tables list the complete power results for all 240 cells together with

the expected combined-outcome contrast $E(\widetilde{Y} \mid R = 1) - E(\widetilde{Y} \mid R = 0)$ used in the code-generated summaries.

Figure 4 shows the sample-size power curves for the remaining two scenarios from the main design, and Figure 5 summarizes the $h = 0$ null-calibration results across all six scenarios.

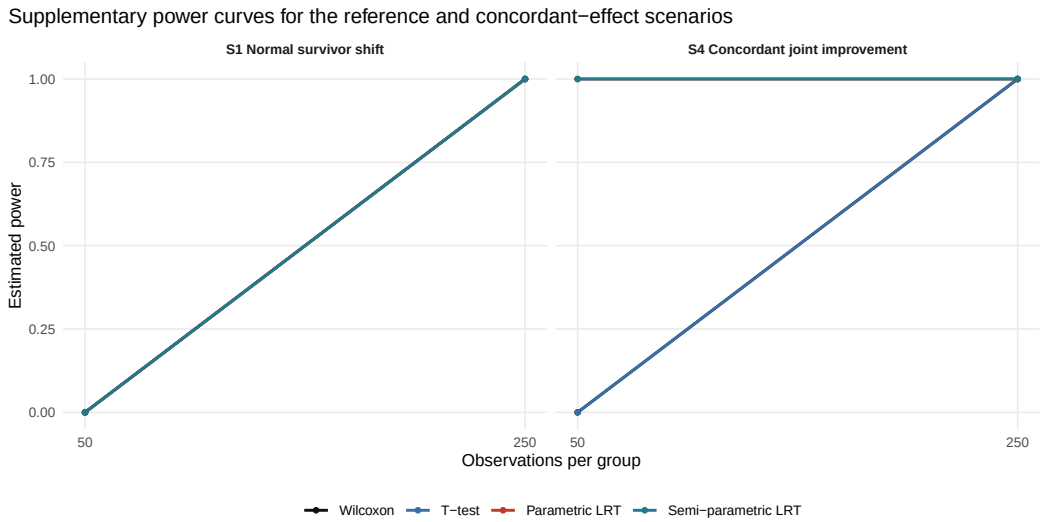

Figure 4: Estimated power as a function of sample size at the strongest non-null level for Scenarios 1 and 4.

Table 2: Normal survivor shift, low atom: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-parametric LRT	Max fail
50	0	$\delta = 0.00$	0.000	0	0	0	0	0
250	0	$\delta = 0.00$	0.000	0	0	0	0	0
50	3	$\delta = 0.50$	0.425	0	0	0	0	0
250	3	$\delta = 0.50$	0.425	100	100	100	100	0

Table 3: Survival-only effect: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-parametric LRT	Max fail
50	0	$\pi_1 = 0.45$	0.00	0	0	0	0	0
250	0	$\pi_1 = 0.45$	0.00	0	0	100	100	0
50	3	$\pi_1 = 0.60$	0.45	100	100	0	0	0
250	3	$\pi_1 = 0.60$	0.45	100	100	100	100	0

Table 4: Antagonistic cancellation: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-pa
50	0	$\pi_1 = 0.50$, $E[Y \mid A = 1, R = 1] = 3.000$	0	0	0	0	

Table 4: Antagonistic cancellation: estimated power by sample size and effect level *(contin*

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-pa
250	0	$\pi_1 = 0.50$, $E[Y \mid A = 1, R = 1] = 3.000$	0	0	0	0	
50	3	$\pi_1 = 0.65$, $E[Y \mid A = 1, R = 1] = 2.308$	0	0	0	100	
250	3	$\pi_1 = 0.65$, $E[Y \mid A = 1, R = 1] = 2.308$	0	100	0	100	

Table 5: Concordant joint improvement: estimated power by sample size and effect leve

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-par
50	0	$\pi_1 = 0.50$, $E[Y \mid A = 1, R = 1] = 3.00$	0.000	0	0	100	
250	0	$\pi_1 = 0.50$, $E[Y \mid A = 1, R = 1] = 3.00$	0.000	0	0	100	
50	3	$\pi_1 = 0.65$, $E[Y \mid A = 1, R = 1] = 3.45$	0.743	0	0	100	
250	3	$\pi_1 = 0.65$, $E[Y \mid A = 1, R = 1] = 3.45$	0.743	100	100	100	

Table 6: Gamma shape-change with mean shift: estimated power by sample size and effect l

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-paran
50	0	$k_1 = 9.0$, $E[Y \mid A = 1, R = 1] = 3.0$	0.00	0	0	0	
250	0	$k_1 = 9.0$, $E[Y \mid A = 1, R = 1] = 3.0$	0.00	0	0	0	
50	3	$k_1 = 2.5$, $E[Y \mid A = 1, R = 1] = 3.6$	0.39	0	0	0	
250	3	$k_1 = 2.5$, $E[Y \mid A = 1, R = 1] = 3.6$	0.39	0	100	100	

Table 7: Rare-outlier contamination: estimated power by sample size and effect level

n	h	Effect	Expected Δ	Wilcoxon	T-test	Parametric LRT	Semi-parametric LRT	Max fail
50	0	$\delta = 0.00$	0.00	0	0	0	0	0
250	0	$\delta = 0.00$	0.00	0	0	0	0	0
50	3	$\delta = 0.35$	0.28	100	0	100	100	0
250	3	$\delta = 0.35$	0.28	100	0	0	0	0

Table 8: Scenario-matched null calibration at effect level $h = 0$

Scenario	n	Wilcoxon	T-test	Parametric LRT	Semi-parametric LRT	Max fail
S1	50	0	0	0	0	0
S1	250	0	0	0	0	0
S2	50	0	0	0	0	0
S2	250	0	0	100	100	0
S3	50	0	0	0	0	0
S3	250	0	0	0	0	0
S4	50	0	0	100	100	0

Table 8: Scenario-matched null calibration at effect level $h = 0$ *(continued)*

Scenario	n	Wilcoxon	T-test	Parametric LRT	Semi-parametric LRT	Max fail
S4	250	0	0	100	100	0
S5	50	0	0	0	0	0
S5	250	0	0	0	0	0
S6	50	0	0	0	0	0
S6	250	0	0	0	0	0

B Supplementary description of the Delta-scale profile and projected intervals

This appendix records the exact unadjusted `TruncComp2` constructions underlying `DeltaProfileCI` and `DeltaProjectedCI`. These intervals are currently implemented only for the no-covariate parametric and semi-parametric models described in Section 2. To unify notation, define

$$\mathcal{W}_n(\mu_\delta, \beta_\delta) = \begin{cases} W_n^P(\mu_\delta, \beta_\delta), & \text{for the parametric fit,} \\ \widetilde{W}_n^P(\mu_\delta, \beta_\delta), & \text{for the semi-parametric fit.} \end{cases}$$

Thus $\mathcal{W}_n$ always denotes the fitted two-parameter joint likelihood-ratio surface on the $(\mu_\delta, \beta_\delta)$ scale.

For the binary component, let

$$\ell_{A,n}(\beta_0, \beta_\delta) = \sum_{i=1}^{n} \left\{ \log\left(1 + e^{\beta_0 + \beta_\delta R_i}\right) - A_i(\beta_0 + \beta_\delta R_i) \right\}$$

denote the negative Bernoulli log-likelihood. For fixed β_δ, define the profiled intercept

$$\widehat{\beta}_0(\beta_\delta) \in \arg\min_{\beta_0 \in \mathbb{R}} \ell_{A,n}(\beta_0, \beta_\delta),$$

and the corresponding fitted observation probabilities

$$\widehat{\pi}_r(\beta_\delta) = \text{logit}^{-1}\left(\widehat{\beta}_0(\beta_\delta) + r\beta_\delta\right), \qquad r \in \{0, 1\}.$$

For the continuous component, let $\widehat{\mu}_0(\mu_\delta)$ denote the profiled control-group mean among the observed outcomes under the fixed mean-difference constraint μ_δ:

$$\widehat{\mu}_0(\mu_\delta) \in \begin{cases} \arg\max_{\mu_0, \Psi} L_{Y,n}(\mu_0, \mu_\delta, \Psi), & \text{parametric,} \\ \arg\min_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta), & \text{semi-parametric,} \end{cases}$$

and define $\widehat{\mu}_1(\mu_\delta) = \widehat{\mu}_0(\mu_\delta) + \mu_\delta$. The profiled combined-outcome contrast induced by the pair $(\mu_\delta, \beta_\delta)$ is then

$$\begin{aligned} g_n(\mu_\delta, \beta_\delta) = {} & \widehat{\pi}_1(\beta_\delta)\widehat{\mu}_1(\mu_\delta) + (1 - \widehat{\pi}_1(\beta_\delta))\mathcal{E} \\ & - \widehat{\pi}_0(\beta_\delta)\widehat{\mu}_0(\mu_\delta) - (1 - \widehat{\pi}_0(\beta_\delta))\mathcal{E}. \end{aligned} \tag{8}$$

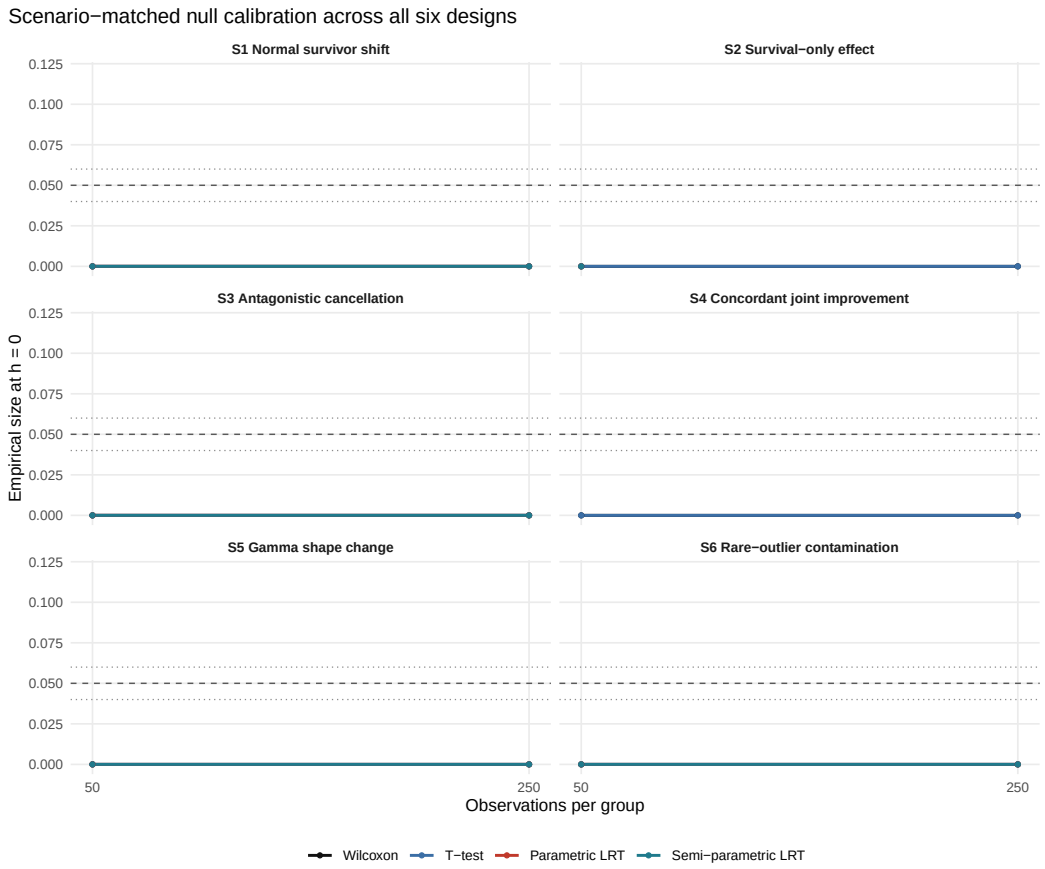

Figure 5: Empirical size at effect level $h = 0$ across all six simulation scenarios. The dashed line marks 0.05 and the dotted lines mark 0.04 and 0.06.

Equation (8) is the map that the software evaluates at every joint candidate point. In particular, the reported point estimate `Delta` is $\widehat{\Delta}_n := g_n(\widehat{\mu}_\delta, \widehat{\beta}_\delta)$.

Let

$$c_{1,\alpha} = \chi_1^2(1-\alpha), \qquad c_{2,\alpha} = \chi_2^2(1-\alpha)$$

denote the one- and two-degree-of-freedom upper quantiles used for the Δ-scale profile and projected intervals, respectively.

Projected interval

The projected interval is the image of the two-dimensional simultaneous confidence region under the map in Equation (8). More precisely, the package reports the endpoint pair

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{proj}} = \left[L_{1-\alpha}^{\mathrm{proj}}, U_{1-\alpha}^{\mathrm{proj}}\right],$$

where

$$L_{1-\alpha}^{\mathrm{proj}} = \inf\left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}, \tag{9}$$

$$U_{1-\alpha}^{\mathrm{proj}} = \sup\left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}. \tag{10}$$

Thus the projected interval is the smallest closed interval containing all Δ values induced by joint parameter pairs inside the $(1-\alpha)100\%$ simultaneous region for $(\mu_\delta, \beta_\delta)$.

For the grid-based approximation used by default, choose a rectangular grid

$$\mathcal{G}_n = \left\{(\mu_\delta^{(a)}, \beta_\delta^{(b)}) : a = 1, \ldots, A,\ b = 1, \ldots, B\right\}.$$

At each grid point define

$$W_{ab} = \mathcal{W}_n(\mu_\delta^{(a)}, \beta_\delta^{(b)}), \qquad g_{ab} = g_n(\mu_\delta^{(a)}, \beta_\delta^{(b)}).$$

The grid estimator of the projected interval is

$$\widehat{\mathrm{CI}}_{1-\alpha}^{\Delta,\mathrm{proj,grid}} = \left[\min\{g_{ab} : W_{ab} \le c_{2,\alpha}\}, \max\{g_{ab} : W_{ab} \le c_{2,\alpha}\}\right],$$

provided that the accepted grid set is non-empty. In the software, the grid bounds are enlarged iteratively until the accepted set no longer touches the boundary of the evaluated box, or until a prespecified maximum number of expansions is reached.

For the optimization-based version, let $B_n \subset \mathbb{R}^2$ be the current rectangular search box on the $(\mu_\delta, \beta_\delta)$ scale. The exact endpoint problems are the constrained programs

$$L_{1-\alpha}^{\mathrm{proj}}(B_n) = \inf_{(\mu_\delta,\beta_\delta)\in B_n} \left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}, \tag{11}$$

$$U_{1-\alpha}^{\mathrm{proj}}(B_n) = \sup_{(\mu_\delta,\beta_\delta)\in B_n} \left\{g_n(\mu_\delta, \beta_\delta) : \mathcal{W}_n(\mu_\delta, \beta_\delta) \le c_{2,\alpha}\right\}. \tag{12}$$

The implementation solves these box-constrained problems numerically by replacing the hard inequality with a one-sided quadratic penalty. Writing $x_+ = \max(x, 0)$ and letting $\kappa_n > 0$ denote the

large data-scaled penalty constant used in the code, the exact objectives passed to the optimizer are

$$Q_{n,\alpha}^{\mathrm{proj},L}(\mu_\delta,\beta_\delta) = g_n(\mu_\delta,\beta_\delta) + \kappa_n\big(\mathcal{W}_n(\mu_\delta,\beta_\delta) - c_{2,\alpha}\big)_+^2, \tag{13}$$

$$Q_{n,\alpha}^{\mathrm{proj},U}(\mu_\delta,\beta_\delta) = -g_n(\mu_\delta,\beta_\delta) + \kappa_n\big(\mathcal{W}_n(\mu_\delta,\beta_\delta) - c_{2,\alpha}\big)_+^2. \tag{14}$$

These expressions are obtained directly from Equations (11)–(12) by taking the lower-endpoint target $g_n(\mu_\delta,\beta_\delta)$ or the upper-endpoint target $-g_n(\mu_\delta,\beta_\delta)$ and then adding a quadratic penalty for positive excess above the simultaneous-region threshold. After numerical optimization of Equation (13) or (14) within B_n, the software moves along the line segment from the fitted center point to the optimizer proposal and performs binary search on that segment to recover the last feasible point satisfying $\mathcal{W}_n \le c_{2,\alpha}$. If either endpoint lies too close to the box boundary, the box is expanded and the search is repeated.

Profile interval

The Δ-profile statistic is the one-dimensional profile of the same joint surface under the nonlinear constraint in Equation (8). For any target value $\Delta \in \mathbb{R}$, define

$$T_n(\Delta) = \inf\left\{\mathcal{W}_n(\mu_\delta,\beta_\delta) : g_n(\mu_\delta,\beta_\delta) = \Delta\right\}. \tag{15}$$

The corresponding profile interval is

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{prof}} = \left\{\Delta \in \mathbb{R} : T_n(\Delta) \le c_{1,\alpha}\right\}.$$

The package reports this set through its lower and upper endpoints,

$$\mathrm{CI}_{1-\alpha}^{\Delta,\mathrm{prof}} = \left[L_{1-\alpha}^{\mathrm{prof}}, U_{1-\alpha}^{\mathrm{prof}}\right],$$

where

$$L_{1-\alpha}^{\mathrm{prof}} = \inf\{\Delta : T_n(\Delta) \le c_{1,\alpha}\}, \qquad U_{1-\alpha}^{\mathrm{prof}} = \sup\{\Delta : T_n(\Delta) \le c_{1,\alpha}\}.$$

Whenever the infimum in Equation (15) is attained on each non-empty fiber $\{(\mu_\delta,\beta_\delta) : g_n(\mu_\delta,\beta_\delta) = \Delta\}$, the accepted Δ values can equivalently be written as

$$\left\{g_n(\mu_\delta,\beta_\delta) : \mathcal{W}_n(\mu_\delta,\beta_\delta) \le c_{1,\alpha}\right\},$$

which explains the grid approximation used in the package. Specifically, on the same joint grid $\mathcal{G}_n$ defined above, the default grid estimator is

$$\widehat{\mathrm{CI}}_{1-\alpha}^{\Delta,\mathrm{prof,grid}} = \left[\min\{g_{ab} : W_{ab} \le c_{1,\alpha}\}, \max\{g_{ab} : W_{ab} \le c_{1,\alpha}\}\right].$$

Thus the profile and projected grid intervals are computed from the same evaluated joint surface and differ only in the threshold: $c_{1,\alpha}$ for the profile interval and $c_{2,\alpha}$ for the projected interval.

For the optimization-based profile interval, the inner profile problem in Equation (15) is reduced to a one-dimensional optimization over β_δ.

In the parametric Gaussian case, write

$$N_+ = \sum_{i=1}^n A_i, \qquad \overline{Y}_+ = \frac{1}{N_+}\sum_{i:A_i=1} Y_i, \qquad \overline{R}_+ = \frac{1}{N_+}\sum_{i:A_i=1} R_i.$$

For fixed μ_δ, the constrained Gaussian fit among observed outcomes minimizes

$$S_n(\mu_0;\mu_\delta) = \sum_{i:A_i=1} (Y_i - \mu_0 - \mu_\delta R_i)^2 .$$

Differentiating with respect to μ_0 and setting the derivative to zero gives

$$\widehat{\mu}_0(\mu_\delta) = \overline{Y}_+ - \overline{R}_+ \mu_\delta, \qquad \widehat{\mu}_1(\mu_\delta) = \overline{Y}_+ + (1-\overline{R}_+)\mu_\delta.$$

Substituting these expressions into Equation (8) yields the exact affine representation

$$\begin{aligned} g_n(\mu_\delta,\beta_\delta) = {} & (\widehat{\pi}_1(\beta_\delta) - \widehat{\pi}_0(\beta_\delta))(\overline{Y}_+ - \mathcal{E}) \\ & + \{\widehat{\pi}_1(\beta_\delta)(1-\overline{R}_+) + \widehat{\pi}_0(\beta_\delta)\overline{R}_+\}\mu_\delta. \end{aligned} \tag{16}$$

Therefore, for fixed target Δ and fixed β_δ, the unique compatible mean-difference value is

$$\mu_\delta^\star(\Delta,\beta_\delta) = \frac{\Delta - (\widehat{\pi}_1(\beta_\delta) - \widehat{\pi}_0(\beta_\delta))(\overline{Y}_+ - \mathcal{E})}{\widehat{\pi}_1(\beta_\delta)(1-\overline{R}_+) + \widehat{\pi}_0(\beta_\delta)\overline{R}_+}, \tag{17}$$

Whenever the denominator is positive, Equation (15) reduces exactly to the one-dimensional problem

$$T_n(\Delta) = \inf_{\beta_\delta\in\mathbb{R}} Q_n^{\mathrm{prof,par}}(\Delta,\beta_\delta), \qquad Q_n^{\mathrm{prof,par}}(\Delta,\beta_\delta) = \mathcal{W}_n(\mu_\delta^\star(\Delta,\beta_\delta),\beta_\delta). \tag{18}$$

In the semi-parametric case, the same reduction is available conceptually but not in closed form. For fixed μ_δ, let

$$\widehat{\mu}_0(\mu_\delta) \in \arg\min_{\mu_0\in\mathbb{R}} \widetilde{W}_{Y,n}(\mu_0,\mu_\delta), \qquad \widehat{\mu}_1(\mu_\delta) = \widehat{\mu}_0(\mu_\delta) + \mu_\delta,$$

so that Equation (8) defines a nonlinear map $(\mu_\delta,\beta_\delta) \mapsto g_n(\mu_\delta,\beta_\delta)$. For fixed (Δ,β_δ), the compatible mean-difference value $\mu_\delta^\star(\Delta,\beta_\delta)$ is any feasible solution of

$$g_n(\mu_\delta,\beta_\delta) = \Delta.$$

Hence the exact profile objective is

$$Q_n^{\mathrm{prof,sp}}(\Delta,\beta_\delta) = \begin{cases} \mathcal{W}_n(\mu_\delta^\star(\Delta,\beta_\delta),\beta_\delta), & \text{if a feasible solution exists,} \\ \infty, & \text{otherwise.} \end{cases} \tag{19}$$

The implementation constructs $\mu_\delta^\star(\Delta,\beta_\delta)$ by searching over the empirical-likelihood feasible range

$$\left[\min_{j\in\mathcal{I}_1} Y_j - \max_{i\in\mathcal{I}_0} Y_i,\ \max_{j\in\mathcal{I}_1} Y_j - \min_{i\in\mathcal{I}_0} Y_i\right].$$

If the function $\mu_\delta \mapsto g_n(\mu_\delta,\beta_\delta) - \Delta$ changes sign over that interval, the code solves the equation by root-finding. Otherwise it minimizes the exact squared-mismatch objective

$$M_n(\mu_\delta;\Delta,\beta_\delta) = \{g_n(\mu_\delta,\beta_\delta) - \Delta\}^2 \tag{20}$$

over the feasible interval and accepts the resulting candidate only if the remaining discrepancy is below numerical tolerance. The outer profile statistic is then

$$T_n(\Delta) = \inf_{\beta_\delta\in\mathbb{R}} Q_n^{\mathrm{prof,sp}}(\Delta,\beta_\delta). \tag{21}$$

Once the profile function $\Delta \mapsto T_n(\Delta)$ has been defined, the lower and upper profile endpoints are the roots of $T_n(\Delta) = c_{1,\alpha}$ on the two sides of the fitted value $\widehat{\Delta}_n$. Numerically, the software brackets these roots by step-doubling away from $\widehat{\Delta}_n$ and then refines the bracket by bisection until the desired tolerance is reached.

C Supplementary derivation of the semi-parametric test statistic

This appendix records the constrained empirical-likelihood calculation underlying Equation (6). Throughout, the treatment-group means among the observed outcomes are parameterized as μ_0 for group $R = 0$ and $\mu_1 = \mu_0 + \mu_\delta$ for group $R = 1$.

Lemma 3. *Fix $(\mu_0, \mu_\delta) \in \mathbb{R}^2$ such that μ_0 lies in the interior of the convex hull of $\{Y_i : i \in \mathcal{I}_0\}$ and $\mu_0 + \mu_\delta$ lies in the interior of the convex hull of $\{Y_j : j \in \mathcal{I}_1\}$. Define the normalized empirical likelihood ratio*

$$R^{\mathrm{E}}_{Y,n}(\mu_0, \mu_\delta) = \sup_{\{p_i\},\{q_j\}} \left\{ \prod_{i \in \mathcal{I}_0} |\mathcal{I}_0| p_i \prod_{j \in \mathcal{I}_1} |\mathcal{I}_1| q_j \right\} \tag{22}$$

subject to

$$\begin{aligned} p_i \geq 0, \quad & \sum_{i \in \mathcal{I}_0} p_i = 1, \qquad & \sum_{i \in \mathcal{I}_0} p_i (Y_i - \mu_0) = 0, \\ q_j \geq 0, \quad & \sum_{j \in \mathcal{I}_1} q_j = 1, \quad & \sum_{j \in \mathcal{I}_1} q_j (Y_j - \mu_0 - \mu_\delta) = 0. \end{aligned}$$

Then the maximizer in Equation (22) is unique and is given by

$$p_i = \frac{1}{|\mathcal{I}_0| \left(1 + \lambda_0 (Y_i - \mu_0)\right)}, \qquad q_j = \frac{1}{|\mathcal{I}_1| \left(1 + \lambda_1 (Y_j - \mu_0 - \mu_\delta)\right)},$$

where (λ_0, λ_1) satisfy

$$\begin{aligned} \frac{1}{|\mathcal{I}_0|} \sum_{i \in \mathcal{I}_0} \frac{Y_i - \mu_0}{1 + \lambda_0 (Y_i - \mu_0)} &= 0, \\ \frac{1}{|\mathcal{I}_1|} \sum_{j \in \mathcal{I}_1} \frac{Y_j - \mu_0 - \mu_\delta}{1 + \lambda_1 (Y_j - \mu_0 - \mu_\delta)} &= 0. \end{aligned}$$

Consequently,

$$\begin{aligned} \log R^{\mathrm{E}}_{Y,n}(\mu_0, \mu_\delta) &= -\sum_{i \in \mathcal{I}_0} \log\left(1 + \lambda_0 (Y_i - \mu_0)\right) - \sum_{j \in \mathcal{I}_1} \log\left(1 + \lambda_1 (Y_j - \mu_0 - \mu_\delta)\right), \\ \widetilde{W}_{Y,n}(\mu_0, \mu_\delta) &= -2 \log R^{\mathrm{E}}_{Y,n}(\mu_0, \mu_\delta) \\ &= 2 \left(\sum_{i \in \mathcal{I}_0} \log\left(1 + \lambda_0 (Y_i - \mu_0)\right) + \sum_{j \in \mathcal{I}_1} \log\left(1 + \lambda_1 (Y_j - \mu_0 - \mu_\delta)\right) \right), \end{aligned}$$

and the profiled statistic is

$$\widetilde{W}^{P}_{Y,n}(\mu_\delta) = \inf_{\mu_0 \in \mathbb{R}} \widetilde{W}_{Y,n}(\mu_0, \mu_\delta).$$

Proof. The objective in Equation (22) is strictly concave on a convex feasible set. The convex-hull assumptions imply that the feasible set contains an interior point, hence the maximizer is unique and satisfies the first-order conditions. Introduce the Lagrangian

$$O(\{p_i\}, \{q_j\}) = \sum_{i \in \mathcal{I}_0} \log(|\mathcal{I}_0| p_i) + \sum_{j \in \mathcal{I}_1} \log(|\mathcal{I}_1| q_j)$$

$$
\begin{aligned}
&+ \gamma_1 \left(\sum_{i \in \mathcal{I}_0} p_i - 1 \right) + \gamma_2 \left(\sum_{j \in \mathcal{I}_1} q_j - 1 \right) \\
&- |\mathcal{I}_0| \lambda_0 \sum_{i \in \mathcal{I}_0} p_i (Y_i - \mu_0) - |\mathcal{I}_1| \lambda_1 \sum_{j \in \mathcal{I}_1} q_j (Y_j - \mu_0 - \mu_\delta). \qquad (23)
\end{aligned}
$$

Differentiation with respect to p_i and q_j yields

$$
0 = \frac{\partial O(\{p_i\}, \{q_j\})}{\partial p_i} = \frac{1}{p_i} - |\mathcal{I}_0| \lambda_0 (Y_i - \mu_0) + \gamma_1, \qquad (24)
$$

$$
0 = \frac{\partial O(\{p_i\}, \{q_j\})}{\partial q_j} = \frac{1}{q_j} - |\mathcal{I}_1| \lambda_1 (Y_j - \mu_0 - \mu_\delta) + \gamma_2. \qquad (25)
$$

Multiplying Equation (24) by p_i and summing over $i \in \mathcal{I}_0$, then using the constraints, gives

$$
\begin{aligned}
0 &= \sum_{i \in \mathcal{I}_0} p_i \frac{\partial O(\{p_i\}, \{q_j\})}{\partial p_i} \\
&= |\mathcal{I}_0| - |\mathcal{I}_0| \lambda_0 \sum_{i \in \mathcal{I}_0} p_i (Y_i - \mu_0) + \gamma_1 \sum_{i \in \mathcal{I}_0} p_i \\
&= |\mathcal{I}_0| + \gamma_1.
\end{aligned}
$$

Hence $\gamma_1 = -|\mathcal{I}_0|$, and the same argument gives $\gamma_2 = -|\mathcal{I}_1|$. Substituting these identities into Equations (24) and (25) yields the stated optimizer weights.

The multiplier equations follow by imposing the moment constraints after substitution of the optimizer. Inserting the optimizer into Equation (22) gives

$$
\begin{aligned}
\log R^{\mathrm{E}}_{Y,n}(\mu_0, \mu_\delta) &= \sum_{i \in \mathcal{I}_0} \log(|\mathcal{I}_0| p_i) + \sum_{j \in \mathcal{I}_1} \log(|\mathcal{I}_1| q_j) \\
&= - \sum_{i \in \mathcal{I}_0} \log\left(1 + \lambda_0 (Y_i - \mu_0)\right) - \sum_{j \in \mathcal{I}_1} \log\left(1 + \lambda_1 (Y_j - \mu_0 - \mu_\delta)\right).
\end{aligned}
$$

The displayed expression for $\widetilde{W}_{Y,n}(\mu_0, \mu_\delta)$ is therefore immediate from the definition $\widetilde{W}_{Y,n}(\mu_0, \mu_\delta) = -2 \log R^{\mathrm{E}}_{Y,n}(\mu_0, \mu_\delta)$, and the profiled statistic is obtained by minimizing over the nuisance mean μ_0. □

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